Where THOUGHT-raising stops and STARTs

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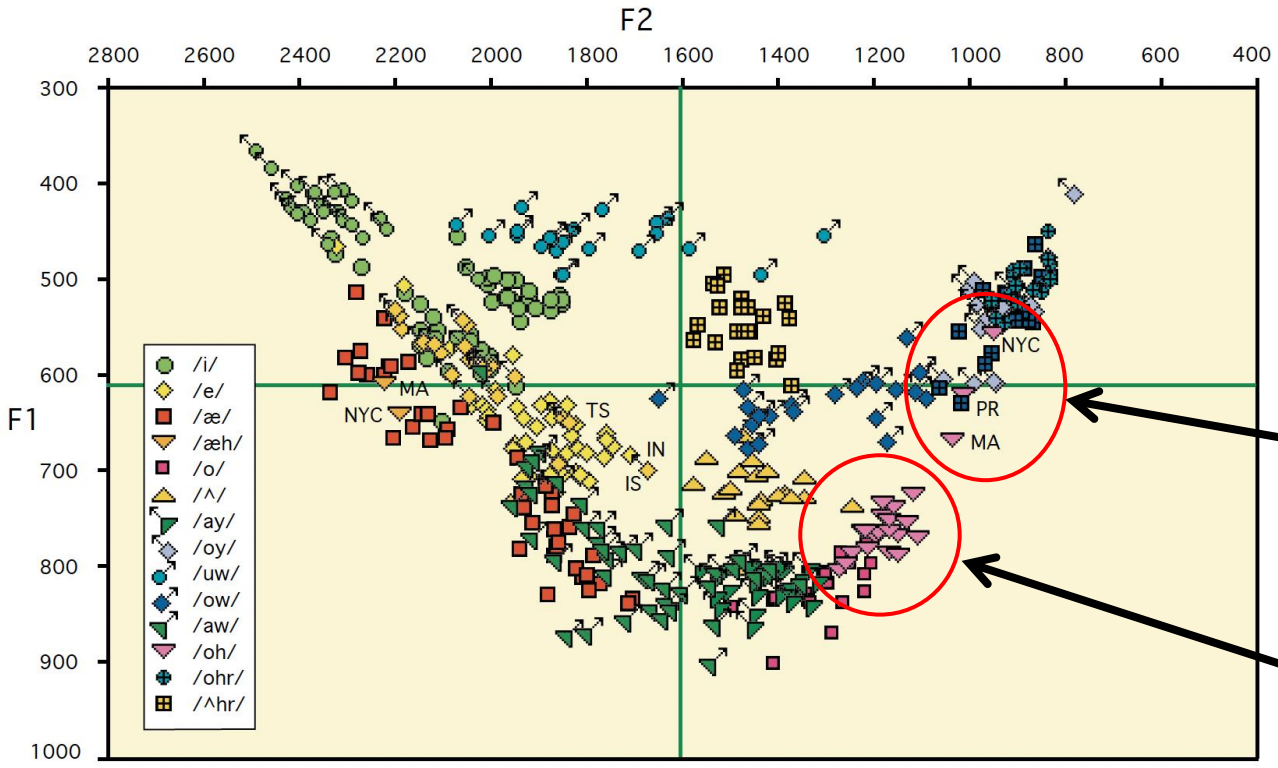
Introduction

- THOUGHT-raising has characterized many dialects of North American English
- However, they differ in the degree of raising
- In this presentation, we answer the following questions:
  1. Why does THOUGHT raise?
  2. How do we account for differences in the degree of THOUGHT-raising?
- Ultimately, we will reconsider the identity of the pre-rhotic back vowels

The alignment of pre-rhotic vowels (Wells 1982)

GOAT	FORCE
THOUGHT	NORTH
PALM	START

Raised THOUGHT in the ANAE



Raised THOUGHT

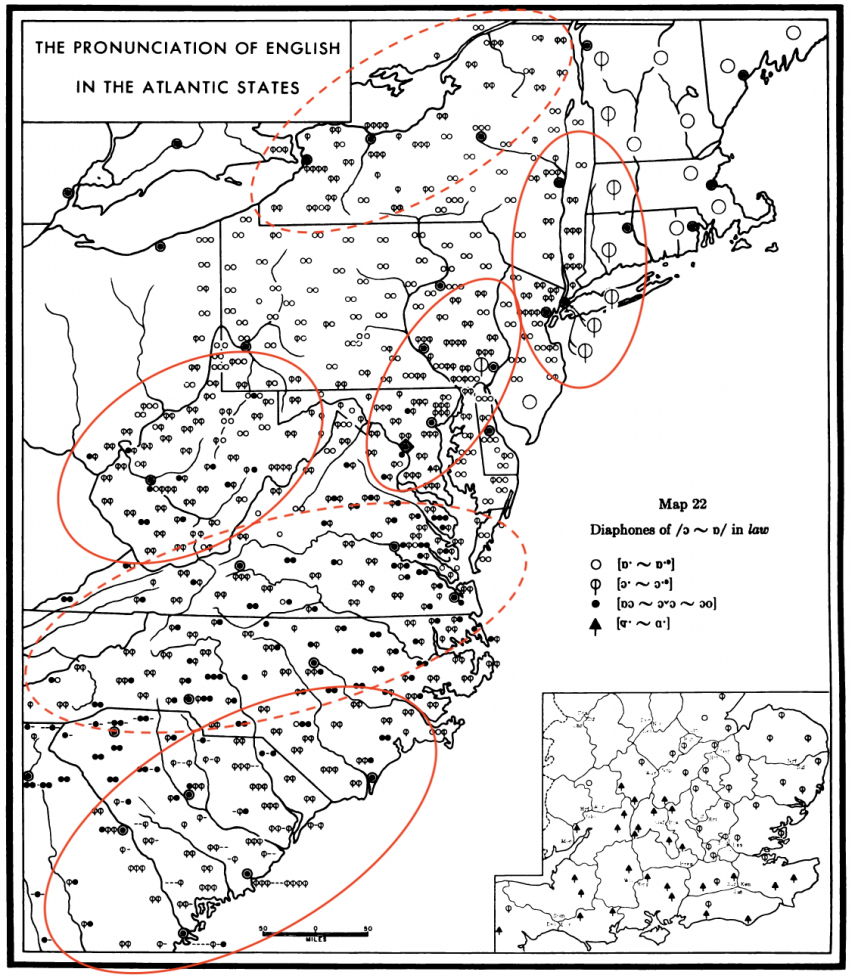
Unraised THOUGHT

Figure 3.5. Peripheral and non-peripheral tracks in the mean file diagram of 14 vowels for the 22 dialects defined in Chapter 11

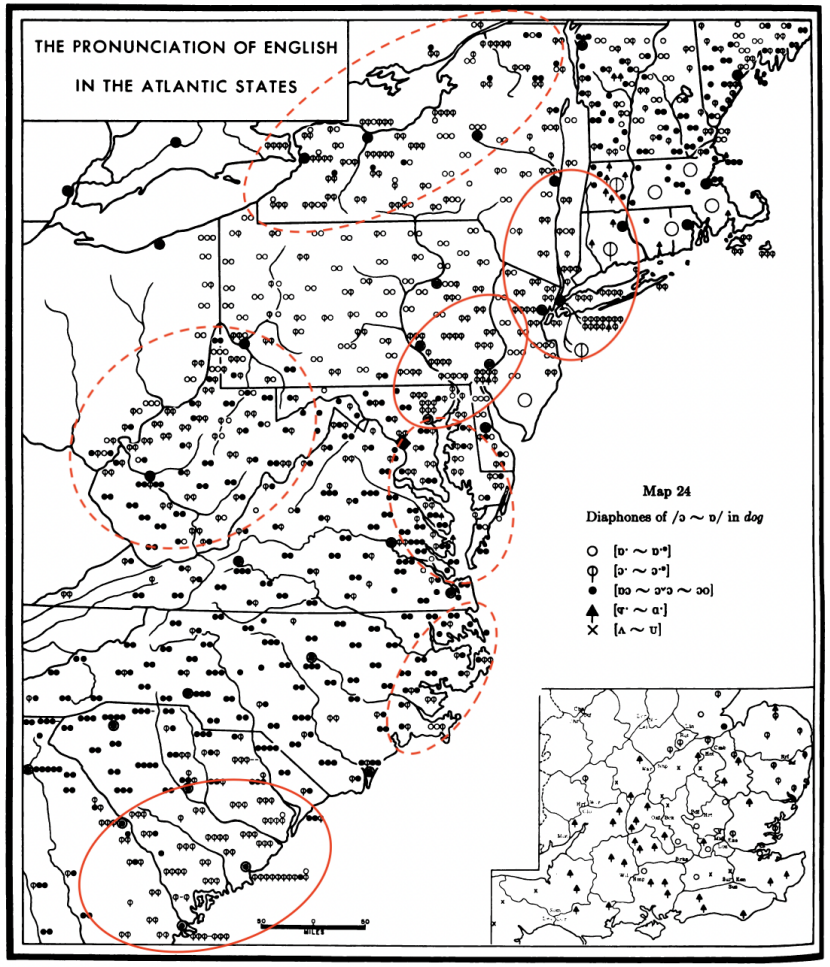
Solid ellipse = raised variant is dominant
Dashed ellipse = raised variant is frequent

Raised THOUGHT in PEAS

Word-final (*law*)

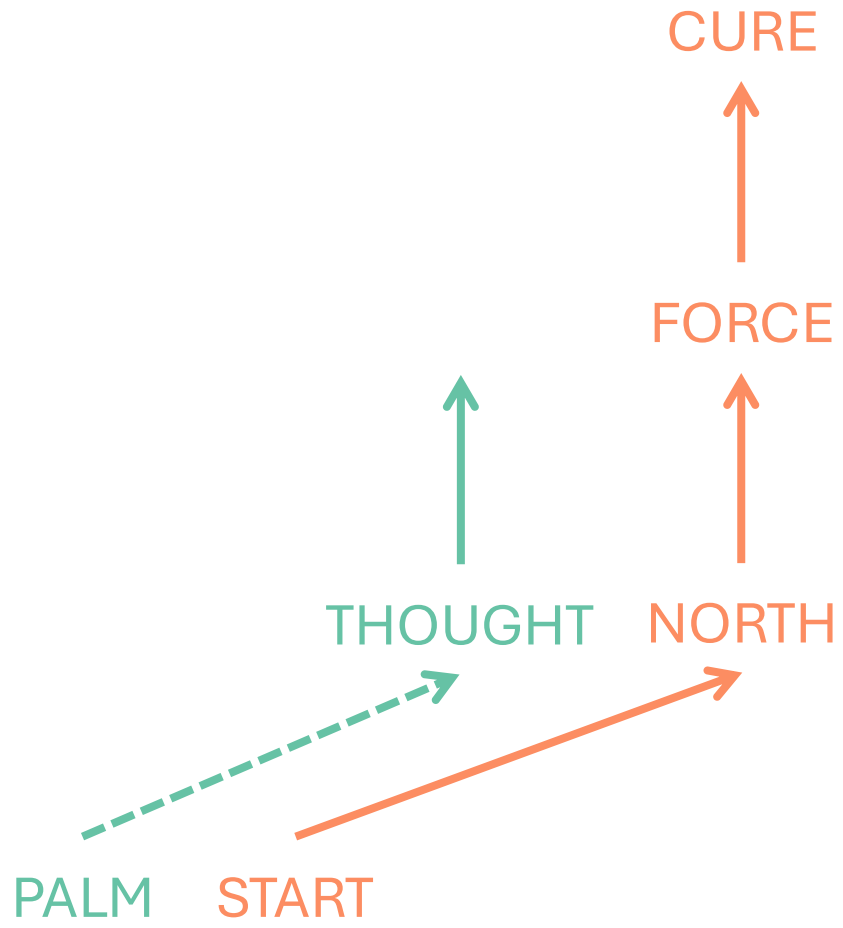


Pre-velar (*dog*)



Back Vowel Chain Shift before /r/ (BVCS)

- Labov et al. (2006) suggest that THOUGHT raises as a generalization of the BVCS

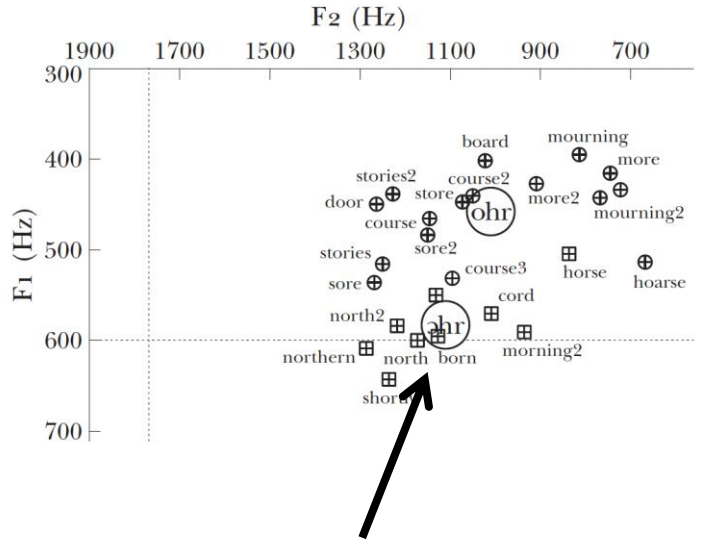


Problems with BCVS analysis

1. THOUGHT-raising occurs without the BVCS
2. Dialects with the BVCS show different degrees of THOUGHT-raising

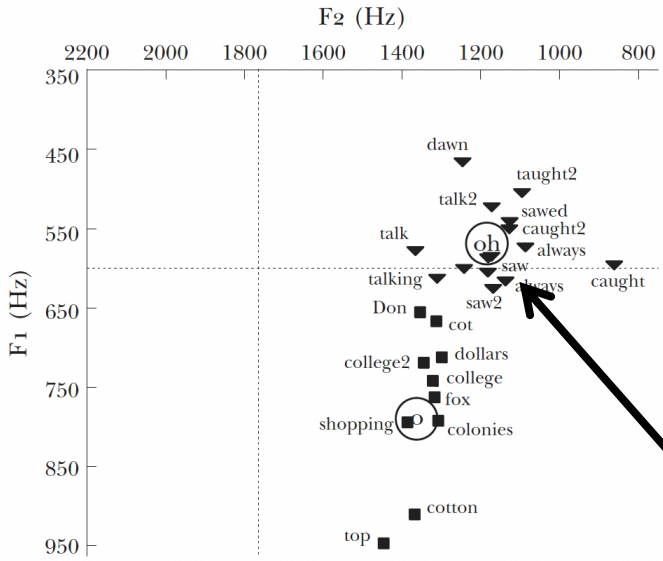
THOUGHT-raising without the BVCS: Charleston (Baranowski 2007)

FIGURE 6.4
Victoria G., 90, Charleston, S.C., Upper-Middle Class: /ɔhr/ and /ohr/



/ɔhr/ = NORTH

FIGURE 7.4
Victoria G., 90, Charleston, S.C., Upper-Middle Class: /o/ and /oh/



/oh/ = THOUGHT

Charleston data from Baranowski 2007. Similar situation holds in Providence (Jaime Benheim, personal communication, 11 November 2025)

THOUGHT-raising without the BVCS

- THOUGHT-raising is a generalization of NORTH-raising (but not the BVCS specifically)

The case of Baltimore

Baltimore Neighborhood Heritage Project (BNHP)

- Oral history interviews conducted 1978–1981 (Shopes 1981)
- Speakers born 1879–1951
- Current sample: 24 speakers born 1896–1940

Race	Gender	Count
Black	F	5
Black	M	3
White	F	8
White	M	8

Corpus of Linguistic Interviews in Charm City (CLICC)

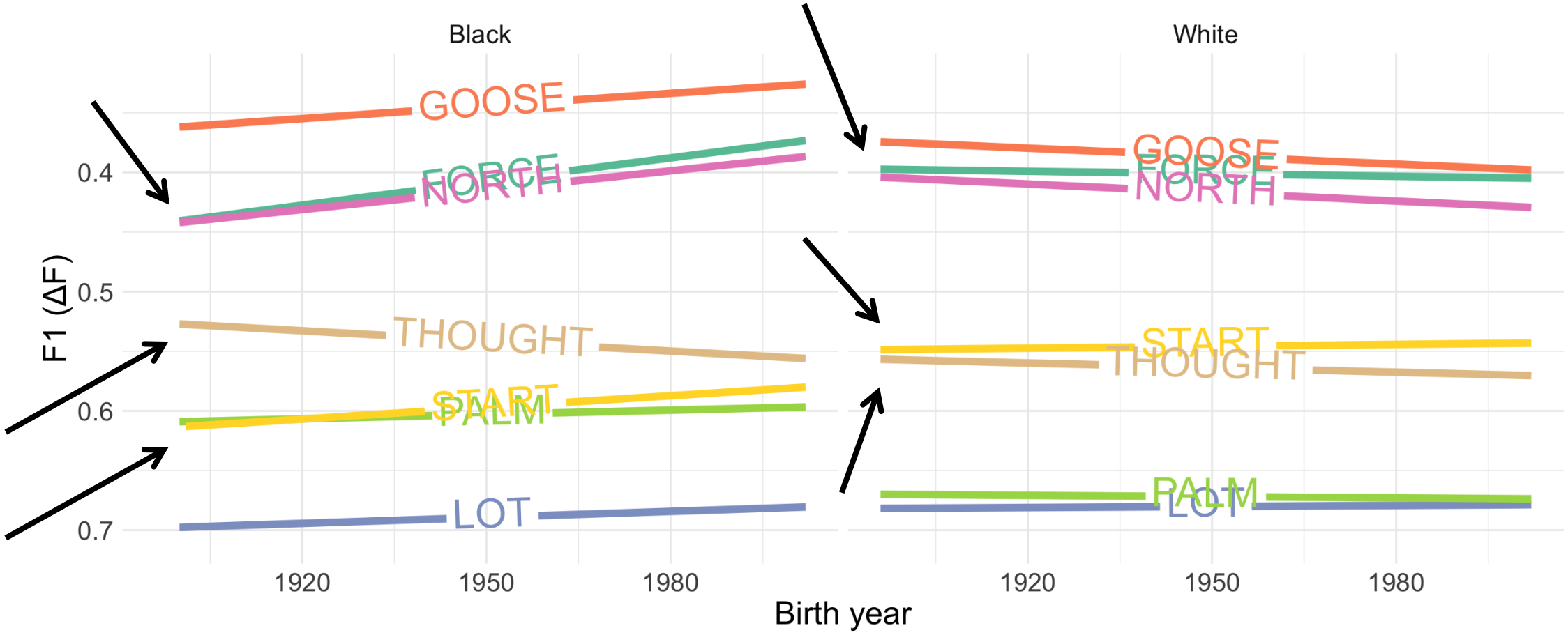
- 48 sociolinguistic interviews conducted 2025–2026
- Elicitation tasks targeting low back vowels and pre-rhotic back vowels
- Speakers born 1940–2008
- Conversational data from 15 participants

Race	Gender	Count (all)	Count (convo.)
Asian	M	1	0
Black	F	8	3
Black	M	6	3
Hispanic	M	1	0
Biracial	F	1	0
Biracial	M	2	0
Biracial	NB	1	0
White	F	15	5
White	M	13	4

Data processing

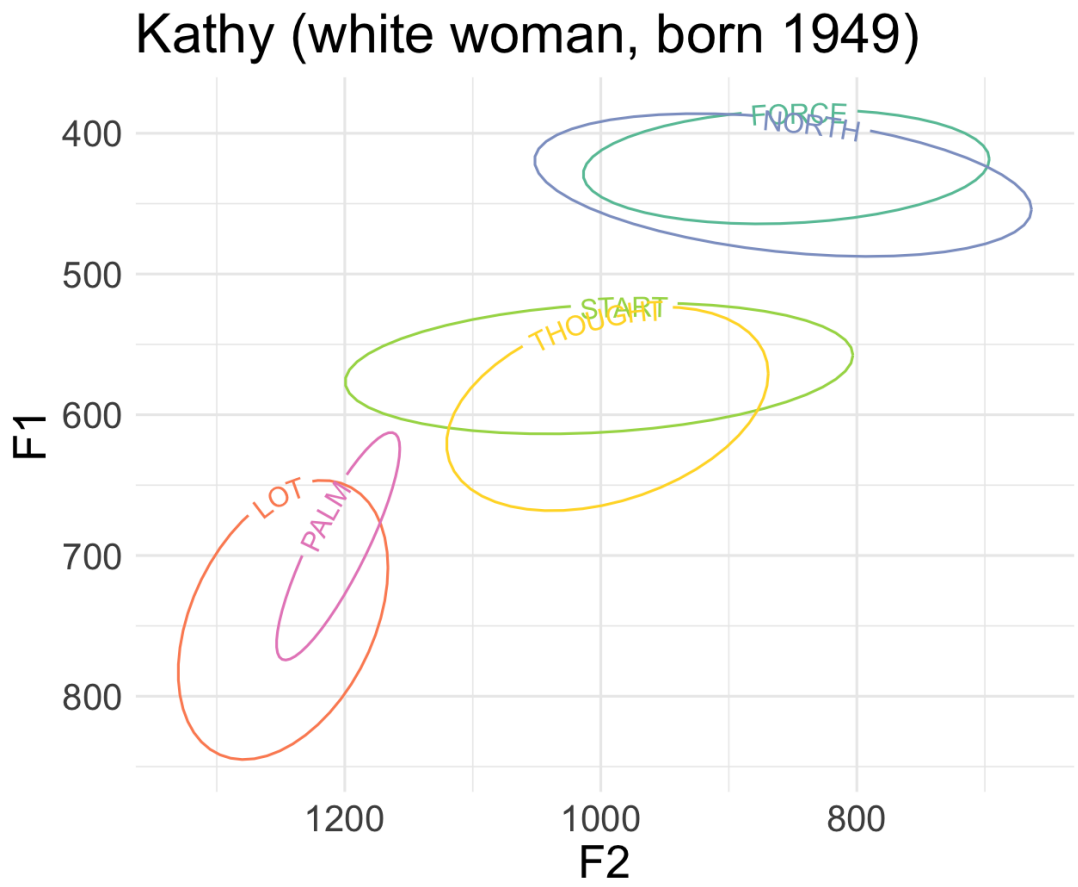
1. Transcription
2. Forced alignment with MFA (McAuliffe et al. 2017)
3. Formant extraction
    1. BNHP: with Darla (Reddy & Stanford 2015)
    2. CLICC: with New-Fave (Fruehwald & Maier 2026)
4. Divide by environment (see Stanley 2022)
5. Outlier removal (Leys et al. 2018)
6. ΔF normalization (Johnson 2020) in Tidynorm (Fruehwald 2025)

THOUGHT never raised past mid position in Baltimore



Lines are robust linear regression smooths

Pre-rhotic THOUGHT is START in Baltimore



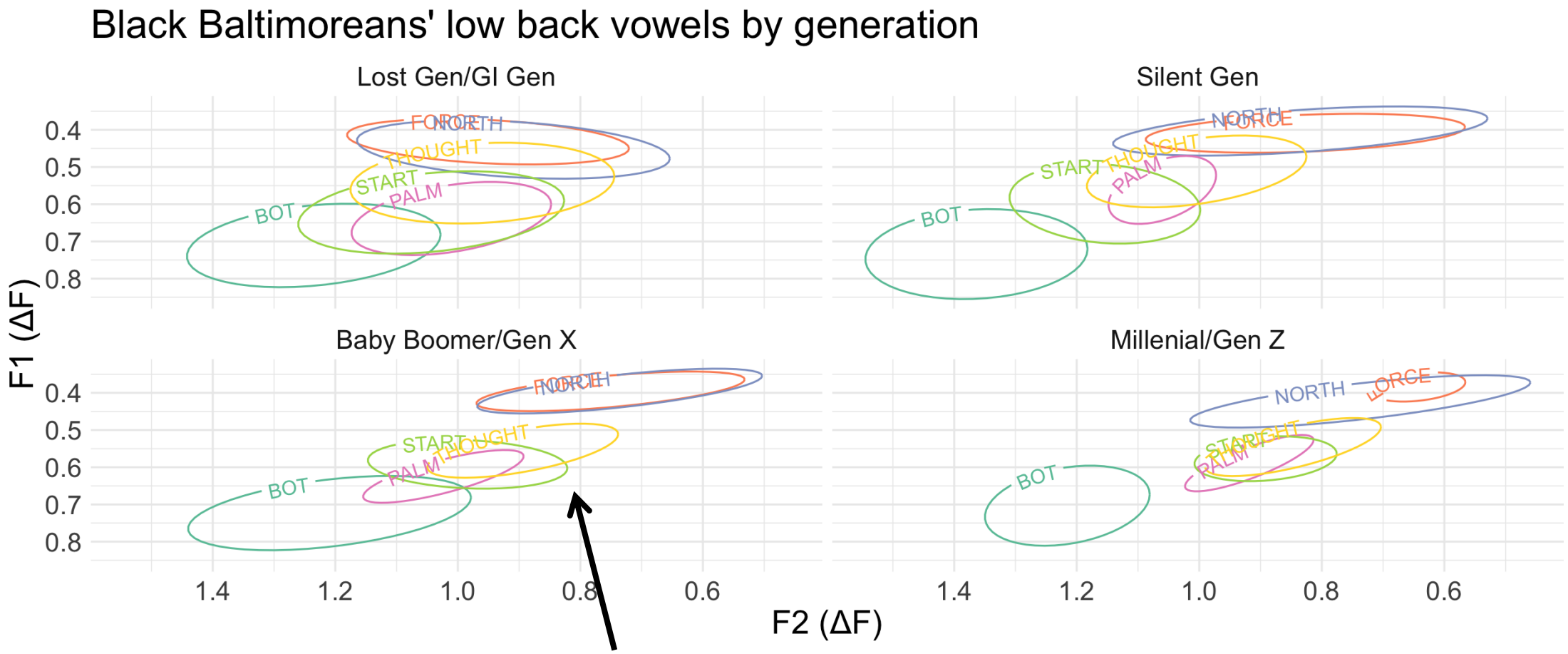
Minimal pair task

Yarn, *yawn*. Well, that sounds a lot (.) alike. Could you not– Yeah, that– that “r” is kinda in the “w.”

The outcome of the BVCS for white Baltimoreans

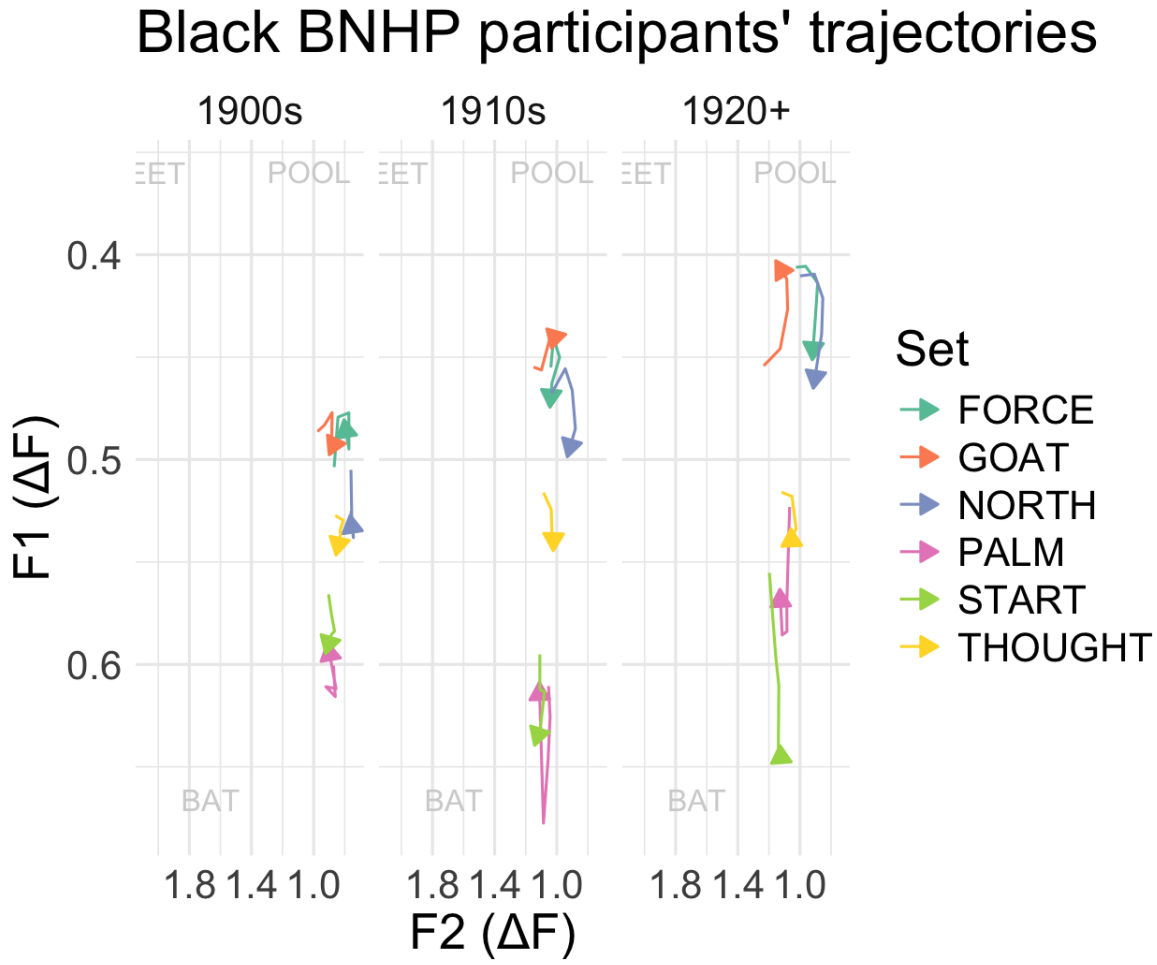
- START replaces NORTH as pre-rhotic THOUGHT
- The precludes the generalization of NORTH-raising to THOUGHT
- THOUGHT is still raised by *ANAE* criteria due to Baltimore's advanced START-raising

START becomes pre-rhotic THOUGHT later for Black Baltimoreans



The outcome of the BVCS for Black Baltimoreans

- For Black Baltimoreans, START becomes pre-rhotic THOUGHT later
- NORTH is likely reanalyzed as pre-rhotic GOAT (via NORTH-FORCE merger) first
- NORTH = pre-rhotic GOAT > NORTH-raising does not generalize to THOUGHT
  - (Or rather, stops generalizing)



Other cities

THOUGHT in New Orleans

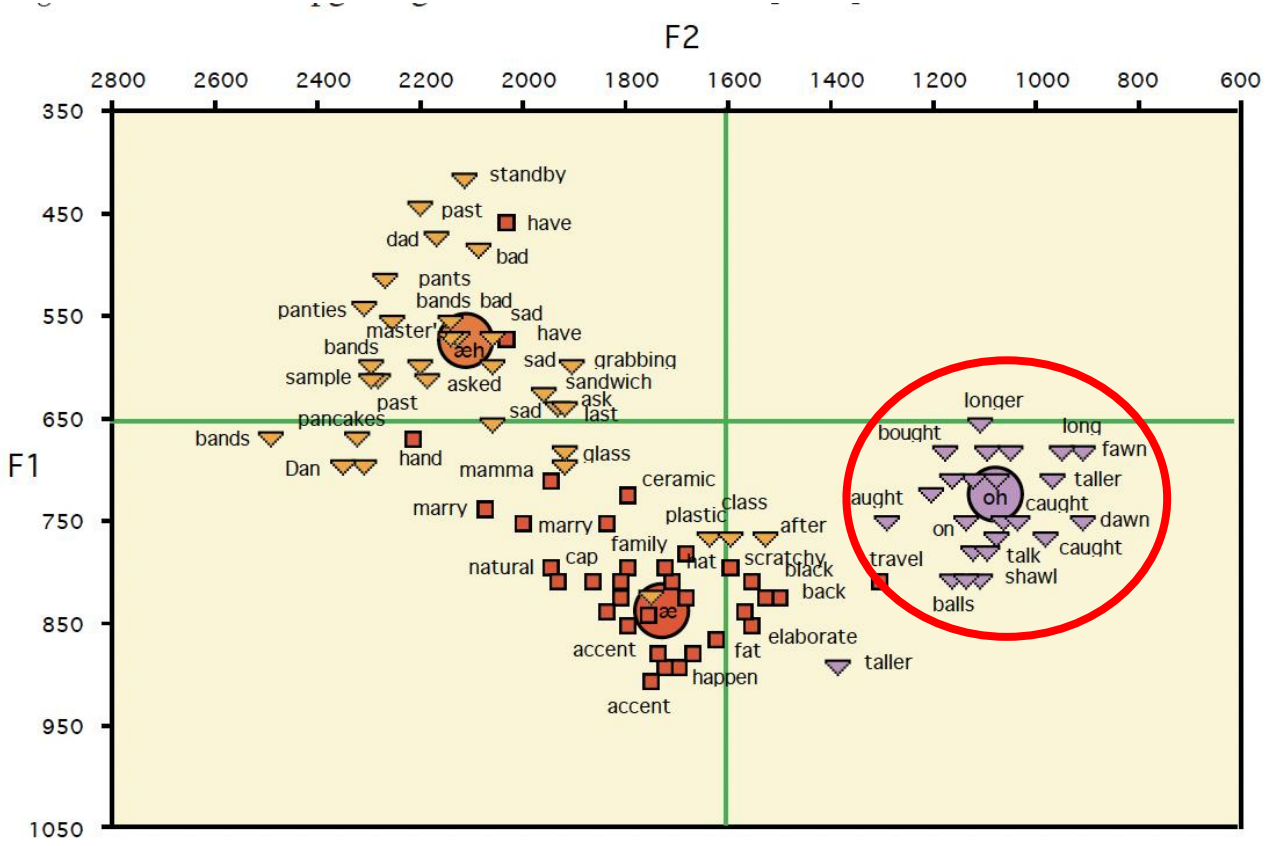


Figure 18.14. Long and ingliding vowels of Edith G., 38 [1996], New Orleans, LA, TS 608

THOUGHT in New Orleans

- New Orleans had a (near) merger of START and NORTH (Dinkin & Carmichael 2025)
- START and THOUGHT (but not NORTH) are now lowering in parallel (Dinkin & Carmichael 2025)
  - Suggests that START is pre-rhotic THOUGHT
- START = pre-rhotic THOUGHT > NORTH-raising does not generalize to THOUGHT

From retreat to reidentification in Philadelphia

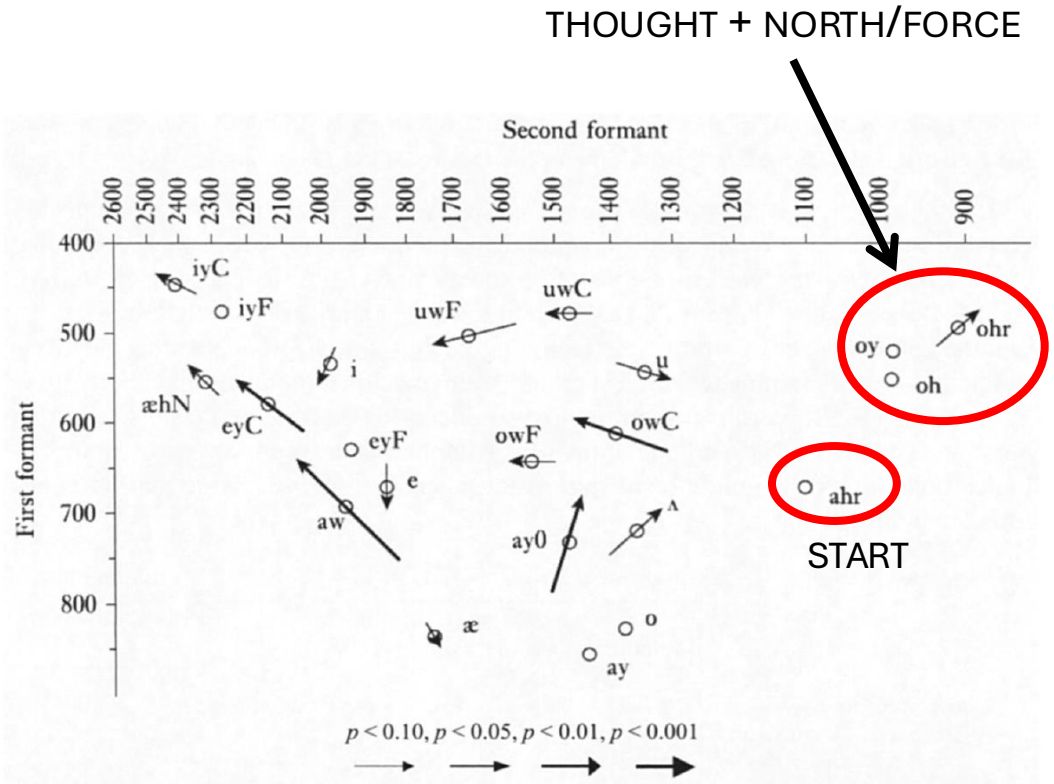


FIGURE 1. Movement of Philadelphia vowels in apparent time of the 1970s. Circles show mean values for 116 speakers in the Philadelphia Neighborhood Study. Heads of arrows show expected values for speakers twenty-five years younger than the mean, tails of arrows for those twenty-five years older than the mean.

Labov et al. (2013)

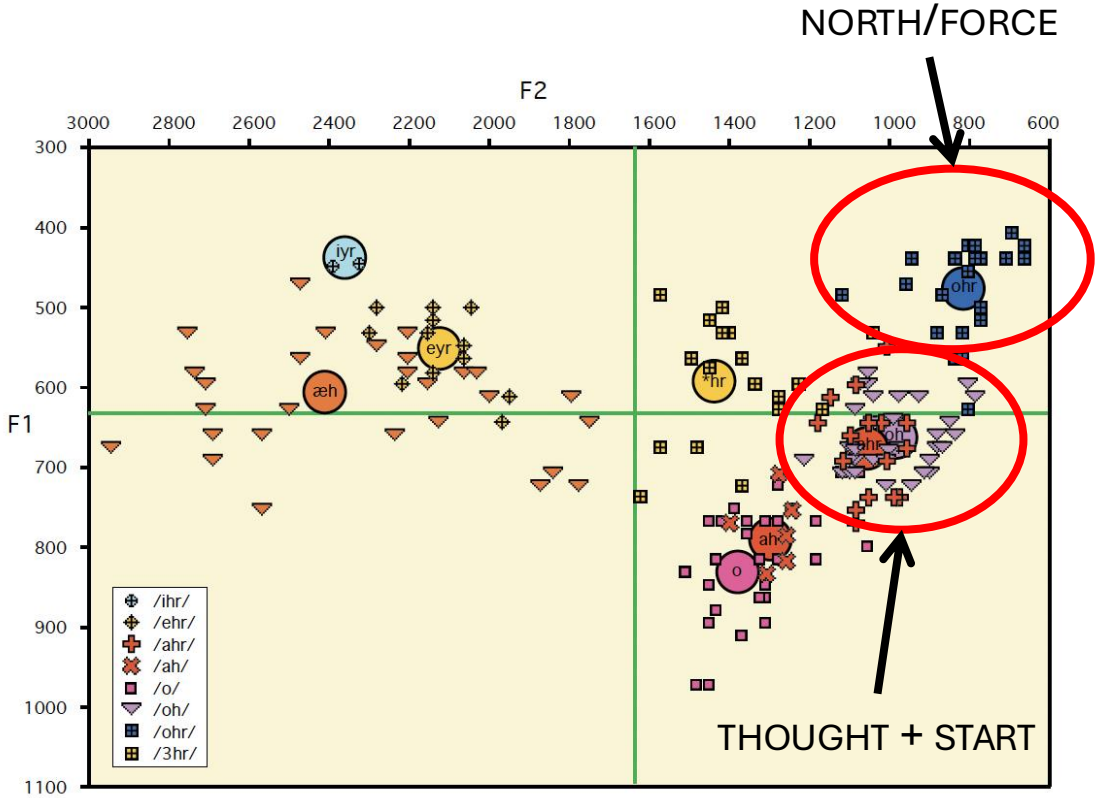


Figure 17.9. Long and ingliding vowels of Rosanne V., 30, Philadelphia, PA, TS 587

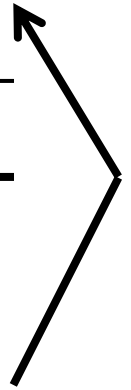
Labov, Ash & Boberg (2006); see also Labov (2010)

Summary and discussion

- THOUGHT raises as a generalization of raising in pre-rhotic sets (but not necessarily NORTH)
- Internal explanation > no diffusion of raised THOUGHT, contra Labov 2007
- The BVCS can lead START to be reanalyzed as pre-rhotic THOUGHT (Labov 2010)—as long as THOUGHT isn't raised with NORTH
  - This describes most of the Southeastern super-region today
  - Possible regional difference between Southeastern region and at least some Northern dialect regions

Alignments of pre-rhotic back vowels

Conservative (e.g., Providence, earlier Charleston)		New York City, earlier Philadelphia		Baltimore, contemporary Southeast	
GOAT	FORCE	THOUGHT	FORCE	GOAT	FORCE
THOUGHT	NORTH		NORTH		NORTH
PALM	START	PALM	START	THOUGHT	START



That's why dissimilatory *r*-lessness turns *forward* into "foe-ward"

Thank you!

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References

Baranowski, Maciej. 2007. *Phonological variation and change in the dialect of Charleston, South Carolina* (Publication of the American Dialect Society 92). Durham, North Carolina: Duke University Press. https://read.dukeupress.edu/pads/issue/92/1.

Dinkin, Aaron J. & Katie Carmichael. 2025. Systems of change: The case of rhotic back vowel systems in New Orleans English. Paper presented at the New Ways of Analyzing Variation, Ann Arbor, Michigan, USA.

Fruehwald, Josef. 2025. tidynorm: Tools for Tidy Vowel Normalization. doi:10.32614/CRAN.package.tidynorm. https://CRAN.R-project.org/package=tidynorm (14 October, 2026).

Josef Fruehwald & Wolfgang Maier. 2026. Forced-Alignment-and-Vowel-Extraction/new-fave: v1.3.0. Zenodo. doi:10.5281/ZENODO.18613417. https://zenodo.org/doi/10.5281/zenodo.18613417 (14 October, 2026).

Kurath, Hans & Raven I. McDavid, Jr. 1961. *The Pronunciation of English in the Atlantic States: Based Upon the Collections of the Linguistic Atlas of the Eastern United States*. University of Michigan Press.

Leys, Christophe, Olivier Klein, Yves Dominicy & Christophe Ley. 2018. Detecting multivariate outliers: Use a robust variant of the Mahalanobis distance. *Journal of Experimental Social Psychology* 74, 150–156. doi:10.1016/j.jesp.2017.09.011.

References

Labov, William, Sharon Ash & Charles Boberg. 2006. *The Atlas of North American English: Phonetics, Phonology, and Sound Change: a Multimedia Reference Tool*. Berlin: Walter de Gruyter.

Labov, William, Ingrid Rosenfelder & Josef Fruehwald. 2013. One Hundred Years of Sound Change in Philadelphia: Linear Incrementation, Reversal, and Reanalysis. *Language* 89(1), 30–65. doi:10.1353/lan.2013.0015.

Labov, William. 2010. *Principles of Linguistic Change, Volume 3: Cognitive and Cultural Factors*. . Vol. 3. 3 vols. John Wiley & Sons. doi:10.1002/9781444327496.

Reddy, Sravana & James Stanford. 2015. A Web Application for Automated Dialect Analysis. In Matt Gerber, Catherine Havasi & Finley Lacatusu (eds.), *Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations*, 71–75. Denver, Colorado: Association for Computational Linguistics. doi:10.3115/v1/N15-3015. https://aclanthology.org/N15-3015/ (14 October, 2026).

Shopes, Linda. 1981. The Baltimore Neighborhood Heritage Project: Oral History and Community Involvement. *Radical History Review* 1981(25), 27–44. doi:10.1215/01636545-1981-25-27.

Stanley, Joseph A. 2022. Interpreting the order of operations in a sociophonetic analysis. *Linguistics Vanguard* 8(1). De Gruyter Mouton, 279–289. doi:10.1515/lingvan-2022-0065.

Wells, John C. 1982. *Accents of English: An introduction*. Vol. 1. Cambridge University Press.

Appendix: START as THOUGHT in Atlanta

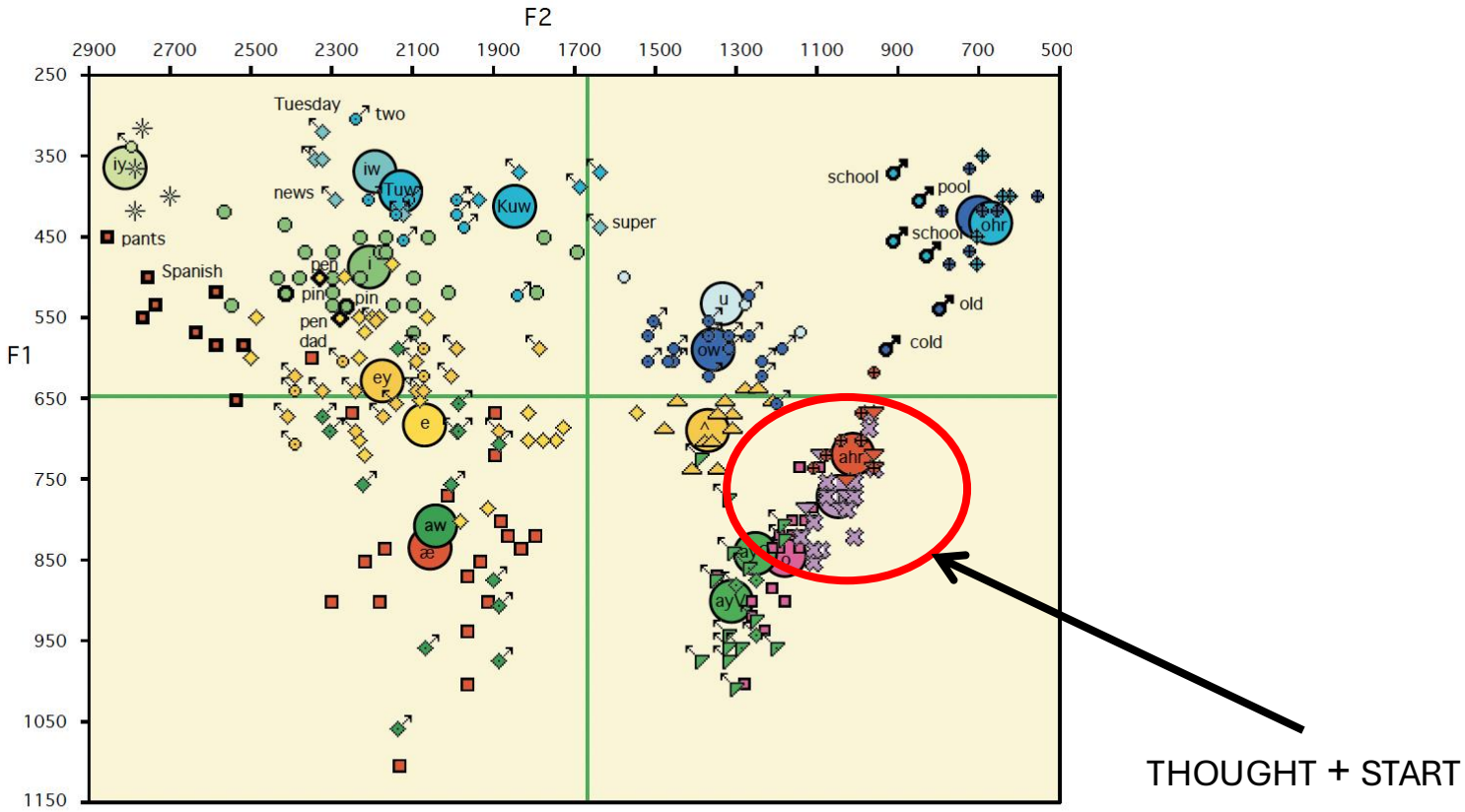


Figure 18.15. Vowel system of Malcolm C., 27 [1999], Atlanta, TS 711. Bold red squares are /æ/ before nasals; bold blue figures are vowels before /l/.